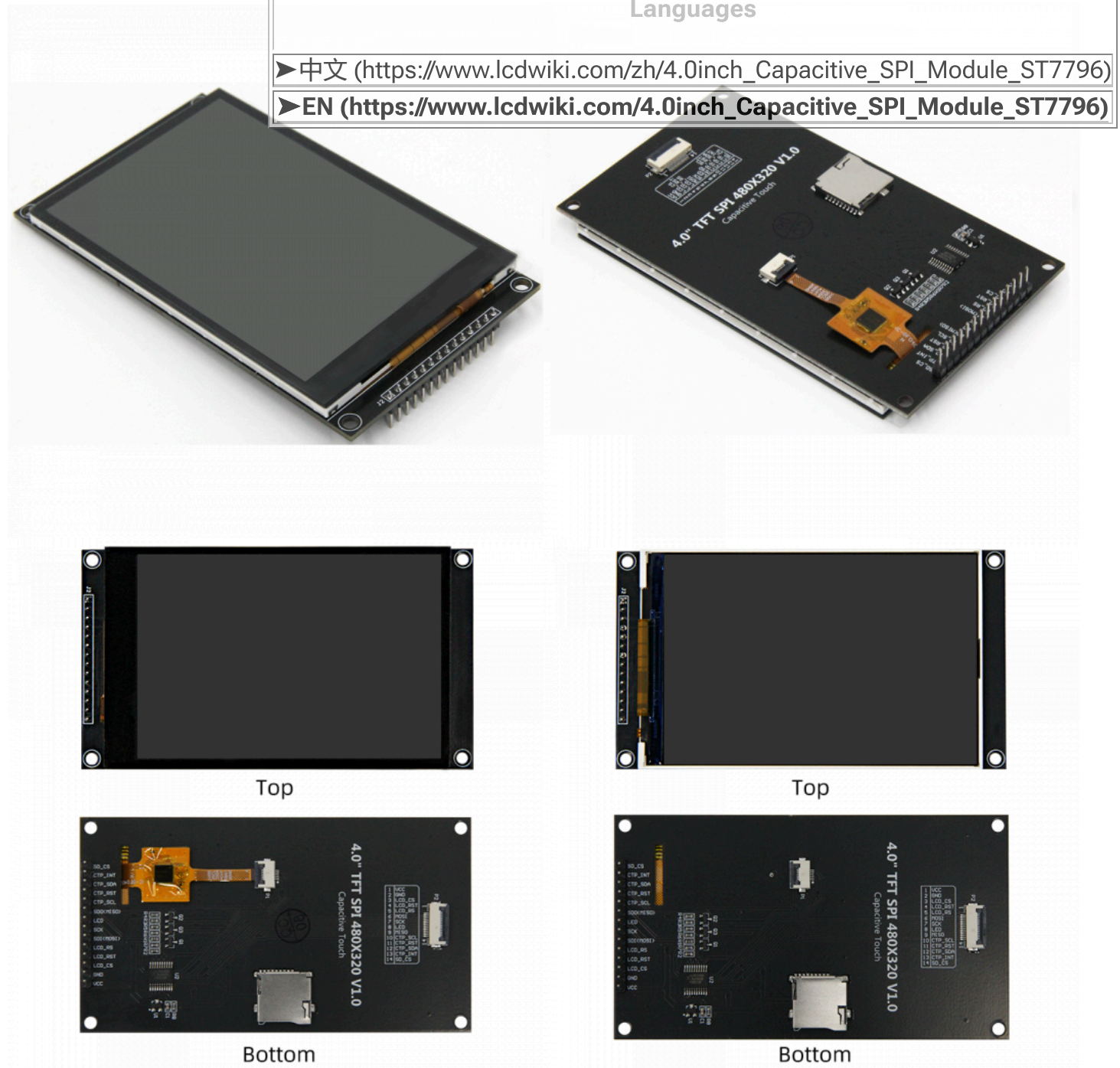


4.0inch Capacitive SPI Module ST7796

From LCD wiki

Product Picture



Product Description

- 4.0inch color screen with 320x480 resolution of 65K colors and rich display colors
- On board level conversion circuit, compatible with 5V and 3.3V MCU
- Adopting a 4-wire SPI serial bus to save I/O pins
- Optional capacitive touch/no touch function for modules
- Module input supports 2.54 pin interface and FPC extension interface
- Equipped with micro TF card slot for easy storage expansion
- Provide rich example learning programs (ESP32/STM32/Arduino UNO&Mage2560/C51/CH32)
- Provide low-level driver technical support, and update WIKI information online
- Multiple tests for module aging can meet military level standards, supporting long-term stable work

► 中文 (https://www.lcdwiki.com/zh/4.0inch_Capacitive_SPI_Module_ST7796)

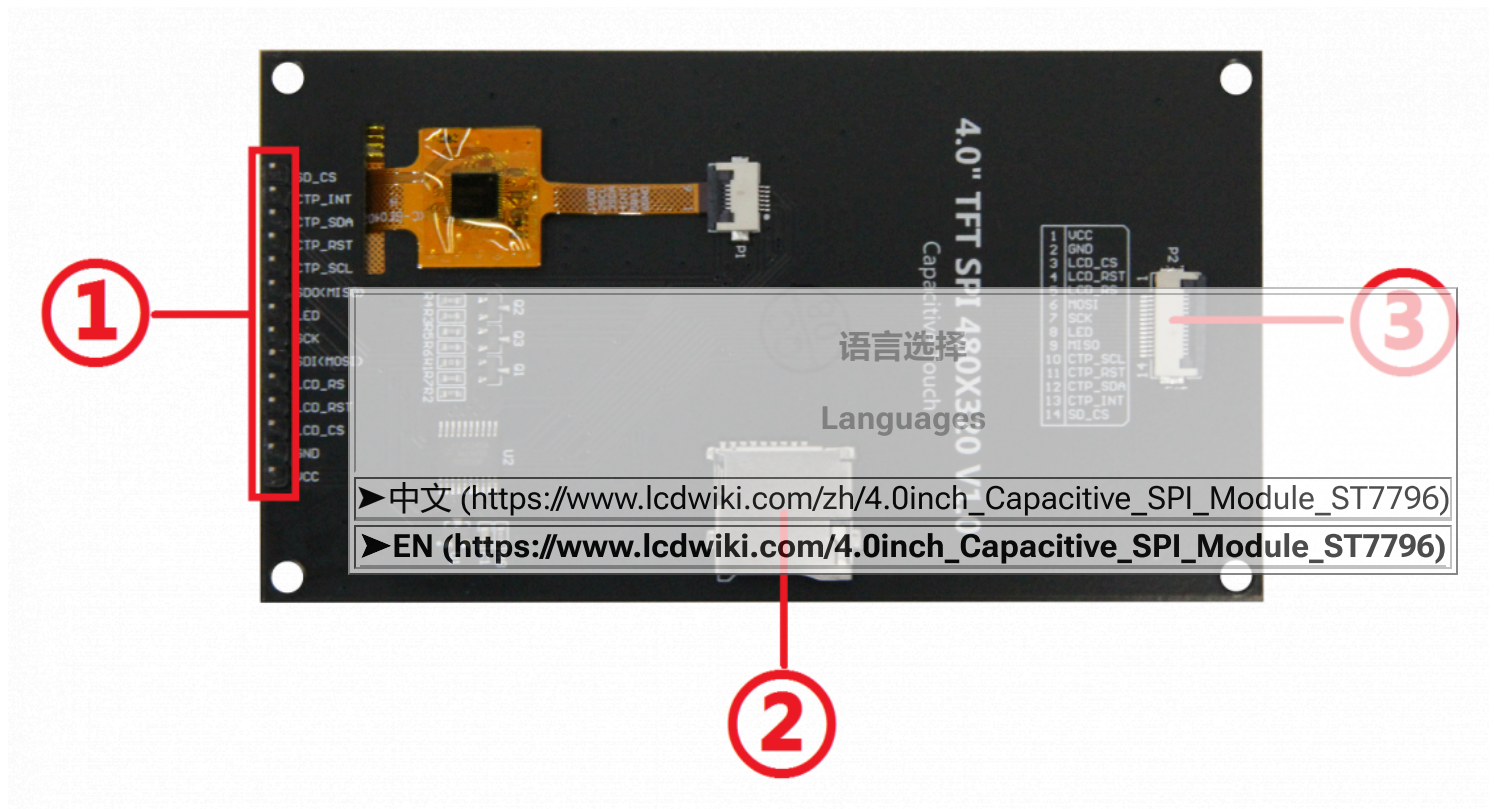
► EN (https://www.lcdwiki.com/4.0inch_Capacitive_SPI_Module_ST7796)

Product Parameters

Name	Parameter
LCD Parameters	
Screen Size	4.0 inch
Screen Type	TN
Resolution	320xRGBx480 (pixels)
Active Area (AA area)	55.68(W)x83.52(H)(mm)
Number of Colors(Max)	16.7M
Driver IC	ST7796S
Display Interface	4-Line SPI
Pixel Size	0.174(H)x0.174(mm)
View Angle	12 O'CLOCK
Brightness(TYP)	300(cd/m ²)
Backlight Type	White LED*8
Operation Temperature	-20~60(°C)
Storage Temperature	-30~70(°C)
Touch Screen Parameters	
Touch Active Area	4.0 inch
Touch Screen Type	Capacitive Touch Screen
Touch Screen Resolution	320x480(pixels)
Driver IC	FT6336U
Touch Screen Visual Area	55.98(W)x83.82(H)(mm)

Communication Interface	IIC
Operation Temperature	-20~70(°C)
Storage Temperature	-30~80(°C)
Size Parameters	
TFT Outline Size	60.88±0.2(W)x94.57±0.2(H)x2.5±0.1(D)(mm)(Excluding cables and adhesive backing)
Touch Screen Outline Size	60.88±0.1(W)x94.57±0.1(H)x1.35 (D)(mm)(Excluding cables and adhesive backing)
Module Outline Size	Have Touch Screen:60.88(W)x108.0(H)x14.80(D)(mm) ► 中文 (https://www.lcdwiki.com/zh/4.0inch_Capacitive_SPI_Module_ST7796) (including Pin Header) ► EN (https://www.lcdwiki.com/4.0inch_Capacitive_SPI_Module_ST7796) NO Touch Screen:60.88(W)x108.0(H)x12.95(D)(mm) (including Pin Header)
Electrical Parameters	
Working Voltage	5.0 V/3.3V(Recommended to connect to 5V)
Backlight Current	103 mA
Power	0.5 W
Other Parameters	
SKU	NO Touch Screen : MSP4030 Have Touch Screen : MSP4031
Interface	14P 2.54mm spacing Header and 0.5mm spacing FPC seat
Weight(including package)	MSP4030: 85g MSP4031: 106g

Interface Definition



• Interface Function Description

Number	Interface	Function Description
①	14 Pin Header	2.54mm spacing row pins, module signal input pins
②	Micro SD Slot	Insert a Micro SD card to expand storage space, such as storing big data content such as fonts and images
③	14P FPC	0.5mm FPC terminal, used for module signal input with the same function as ①

• Pin Function Description

Number	Pin Label	Description
1	VCC	Power positive(It is recommended to connect to 5V. When connected to 3.3V, the backlight brightness will be slightly dim)
2	GND	Power ground
3	LCD_CS	LCD selection control signal, Low level active
4	LCD_RST	LCD reset control signal, Low level reset
5	LCD_RS	LCD command / data selection control signal High level: data, low level: command

6	SDI(MOSI)	SPI bus write data signal(SD card and LCD screen used together)
7	SCK	SPI bus clock signal(SD card and LCD screen used together)
8	LED	LCD backlight control signal (If you need control, please connect the pins. If you don't need control, you can skip it)
9	SDO(MISO)	SPI bus read data signal (SD card and LCD screen used together)
10	CTP_SCL	Capacitive touch screen IIC bus clock signal (modules without touch screens do not need to be connected)
11	CTP_RST	Capacitor touch screen reset control signal, low-level reset (modules without touch screens do not need to be connected)
12	CTP_SDA	Capacitive touch screen IIC bus data signal (modules without touch screens do not need to be connected)
13	CTP_INT	Capacitor touch screen IIC bus touch interrupt signal, when generating touch, input low level to the main control (modules without touch screens do not need to be connected)
14	SD_CS	SD card selection control signal, low level active (without SD card function, can be disconnected)

Product Documentation

- **4.0inch SPI Module Specification** (https://www.lcdwiki.com/res/MSP4030_MSP4031/MSP4030_MS_P4031_Specification_EN_V1.0.pdf)
- **4.0inch SPI Module User Manual** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_Module_MSP4030_MSP4031_User_Manual_EN.pdf)
- **ST7796 initialization** (https://www.lcdwiki.com/res/MSP4030_MSP4031/ST7796_Init.txt)
- **4.0inch LCD Specifications** (https://www.lcdwiki.com/res/MSP4030_MSP4031/QD3958-Specifications-A.pdf)
- **4.0inch LCD Size** (https://www.lcdwiki.com/res/MSP4030_MSP4031/QD3958_Size.pdf)
- **4.0inch Capacitive Touch Screen Size Picture** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_CTP_Module_FT6336U_Size.pdf)
- **4.0inch SPI Module with Touch Screen Size Picture** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_Module_MSP4031_Size.pdf)
- **4.0inch SPI Module without Touch Screen Size Picture** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_Module_MSP4030_Size.pdf)
- **4.0inch SPI Module Schematic** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_Module_MSP4030_MSP4031_Schematic.pdf)
- **4.0inch SPI Module screen Schematic and PCB package library** (https://www.lcdwiki.com/res/MSP4030_MSP4031/Altium_4.0inch_48pin_QD3958_Package_library.zip)
- **LCD Driver IC ST7796S Data sheet** (https://www.lcdwiki.com/res/MSP4030_MSP4031/ST7796S-Sitronix.pdf)
- **Touch Screen Driver IC FT6336U Data sheet** (https://www.lcdwiki.com/res/MSP4030_MSP4031/DF_T6336UDataSheetV1.1.pdf)
- **Touch Screen Driver IC FT6336U Register Table** (https://www.lcdwiki.com/res/MSP4030_MSP4031/FT6336U_Register.xlsx)

Program Download

- **4.0inch SPI Module MSP4030_MSP4031 Package** (https://www.lcdwiki.com/res/Program/CTP_SPI/4.0inch/SPI_ST7796_MSP4030_MSP4031_V1.0/4.0inch_SPI_Module_ST7796_MSP4030_MSP4031_V1.0.zip)

Reference Materials

语言选择

Languages

- **4.0inch SPI Module C51 Demo Instructions** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_MSP4030_MSP4031_C51_Demo_Instructions_EN.pdf)
- **4.0inch SPI Module C51 Demo Instructions** (https://www.lcdwiki.com/zh/4.0inch_Capacitive_SPI_Module_ST7796/4.0inch_SPI_MSP4030_MSP4031_C51_Demo_Instructions_CN.pdf)
- **4.0inch SPI Module C51 Demo Instructions** (https://www.lcdwiki.com/4.0inch_Capacitive_SPI_Module_ST7796/4.0inch_SPI_MSP4030_MSP4031_C51_Demo_Instructions_EN.pdf)
- **4.0inch SPI Module ESP32 Demo Instructions** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_MSP4030_MSP4031_ESP32_Demo_Instructions_EN.pdf)
- **4.0inch SPI Module STM32 Demo Instructions** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_MSP4030_MSP4031_STM32_Demo_Instructions_EN.pdf)
- **4.0inch SPI Module UNO_Mega2560 Demo Instructions** (https://www.lcdwiki.com/res/MSP4030_MSP4031/4.0inch_SPI_MSP4030_MSP4031_UNO_Mega2560_Demo_Instructions_EN.pdf)
- **Arduino development environment construction for ESP32** (https://www.lcdwiki.com/res/PublicFile/Arduino_development_environment_construction_for_ESP32_EN.pdf)
- **PCtoLCD2002 software use illustration** (https://www.lcdwiki.com/res/PublicFile/PCtoLCD2002_Use_Illustration_EN.pdf)
- **Image2Lcd software use illustration** (https://www.lcdwiki.com/res/PublicFile/Image2Lcd_Use_Illustration_EN.pdf)
- **flash_download_tool software use illustration(use for ESP32)** (https://www.lcdwiki.com/res/PublicFile/Flash_Download_Operation_Instructions_EN.pdf)
- **Chinese and English display modulo settings** (https://www.lcdwiki.com/Chinese_and_English_display_modulo_settings)

Common Software

- **PCtoLCD2002** (<https://www.lcdwiki.com/res/software/PCtoLCD2002.zip>)
- **Image2Lcd** (<https://www.lcdwiki.com/res/software/Image2Lcd.zip>)
- **flash_download_tool** (https://www.lcdwiki.com/res/software/flash_download_tool_3.9.4.zip)

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